Emotion Recognition for Live Camera Feed

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***Abstract*— Emotion recognition is the ability to identify and classify human emotions. It has gained significant traction in recent years due to its potential applications in various domains, including human-computer interaction, affective computing, and security surveillance. One promising technique for emotion recognition is real-time emotion detection from live camera feeds. This technique can capture spontaneous and dynamic emotional expressions.**

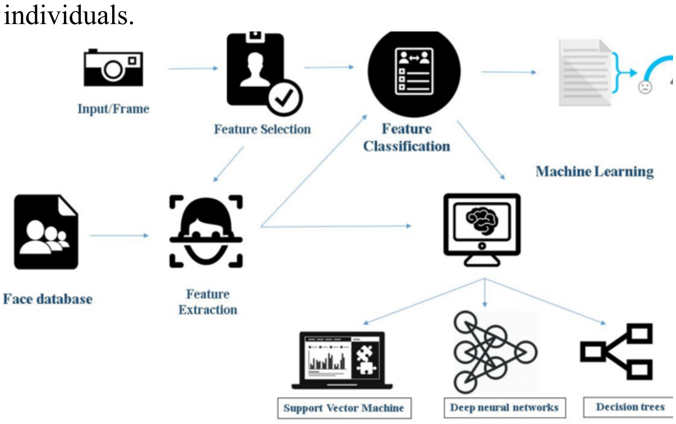
This paper proposes a prototype system for emotion/facial expression recognition using a neural network and image processing. The system can classify six universal emotions: happiness, sadness, anger, disgust, surprise, fear, and neutral expression. It takes live camera feed from the webcam of the device and extracts feature points from the face(s) recognized in the frames obtained. These feature points are then used to train a neural network to recognize the emotions.

***Keywords* -** *real time emotion analysis, facial expression recognition, image processing, convolutional neural network*

I. Introduction

THE primary nonverbal cue through which humans can express their emotions is their facial expression. Facial expression accounts for 55% of total videographic communication while vocal and spoken communication contributes 38% and 7%, respectively. [1] [3]

These facial expressions can convey six basic emotions: happiness, sadness, anger, fear, surprise, disgust and neutral. While humans can instantly recognize these expressions, automated systems require advanced algorithms and adequate computation power to handle variations in facial expressions across cultures and individuals.



Facial expression recognition (FER) is an emerging field that aims to address this challenge. FER systems can be used in a variety of applications, including Human Computer Interaction (HCI), Affective Computing, VR gaming, Social Media Reactions, and many more.

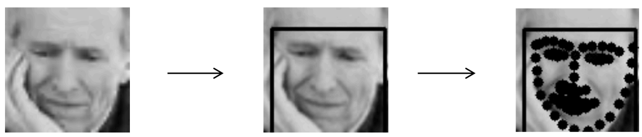
FER systems work by analyzing facial features and expressions in images or videos. FER systems can also be classified based on the type of features they use. [4] [5] Geometric features are based on the position of facial landmarks, while appearance features are based on the texture of the skin. [1]

They use machine learning algorithms to identify and categorize these expressions. Convolutional neural networks (CNNs) are a type of machine learning algorithm that is particularly well-suited for this task.

Here, we process data feed directly from device webcam to analyze emotion/facial expression using algorithm discussed later in this report.

The motivation for this work is the need for an algorithm that can automatically identify and classify facial expressions in real time. Such an algorithm would have a wide range of applications as stated earlier.

II. Steps Involved in Emotion Recognition



(Input: Sad Face; Face and features are subsequently extracted)

1. ***Face Detection***
   Face Detection involves identifying human faces in images, addressing challenges like variations in size, angle, and poses. Past approaches include knowledge-based, feature invariant, template-based, and appearance-based methods. Geometric Analysis and sample datasets are crucial for Face Detection.
   Face Detection is vital for Emotion Recognition (ER) systems, facilitating the analysis of facial expressions and emotional cues in images or videos. ER systems rely on accurate face detection to produce desired results.

2. *Feature Extraction*

During the feature extraction phase, the face identified in the preceding step undergoes additional processing to recognize the eye, eyebrow, and mouth regions. Initially, the probable Y coordinates of the eyes are determined through horizontal projection. Subsequently, the regions around these Y coordinates are examined to pinpoint the precise locations of the features. Ultimately, a corner point detection algorithm is employed to acquire the necessary corner points from the identified feature regions.

i. *Eye Extraction*

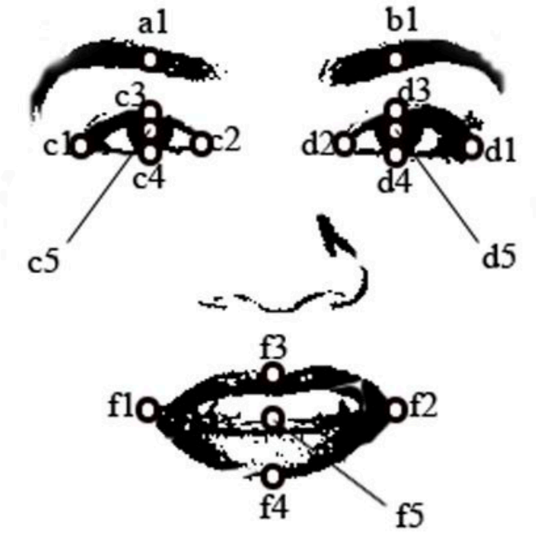
The method involves detecting the Y coordinate of the eyes by applying the Sobel mask to an image, generating a horizontal projection of vertical edges. Using a single Sobel mask was insufficient in accurately determining the Y coordinate, so Sobel edge detection is applied exclusively to the upper half of the facial image. The horizontal sum of each row is plotted, and the top two peaks in the horizontal projection of edges are identified. The peak with the lower intensity value in the horizontal projection of intensity is chosen as the Y coordinate of the eyes. The region around the Y coordinate is processed to identify areas satisfying specific geometric criteria and the condition G < 60, with G representing the green color component in the RGB color space. These regions are refined using the Roberts method for edge detection, followed by dilation and hole filling. Expansion of regions continues until they reach the edges, resulting in the final identification of eye regions based on specific geometric conditions. [2, 18-24]

ii. *Eyebrow Extraction*

To identify the eyebrow regions, two rectangular areas located directly above each eye region in the edge image are selected. Edge images for these two areas are generated to further enhance their definition. The Sobel method is used for edge detection due to its superior edge-detection capabilities compared to the Roberts method. These generated edge images are then dilated, and any existing holes are filled. These edge images are further used to identify the eyebrow regions.

iii. *Mouth Extraction*

In the image processing procedure, the area below the eyes is analyzed to identify regions meeting a specific condition: (1.2 ≤ R/G ≤ 1.5), R & G are standard values as used in RGB representation of images. Simultaneously, the edge image for this area is generated using the Roberts method, followed by dilation and hole filling. A refined mouth region is selected from the regions satisfying the condition, and further refinement involves expanding this region until it includes the edges in the processed image. After identifying feature regions, such as eyes and mouth, additional processing is carried out to extract key feature points. The Harris corner point detection algorithm is utilized to locate the leftmost and rightmost corner points of the eyes and mouth. Midpoints between these corner points, along with additional information, aid in calculating top, bottom, rightmost, and leftmost points and centroids for both the eyes and mouth. Additionally, a point in the eyebrow directly above the eye center is determined through the analysis of the edge image.



Primary Features for Emotion Recognition

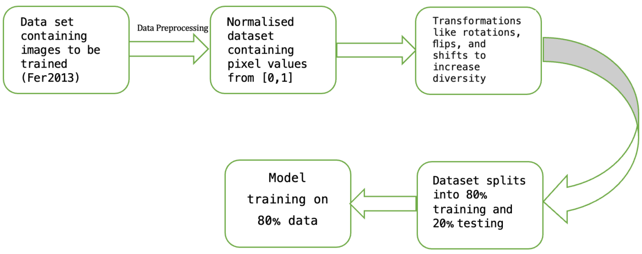
3. *Emotion Classification*

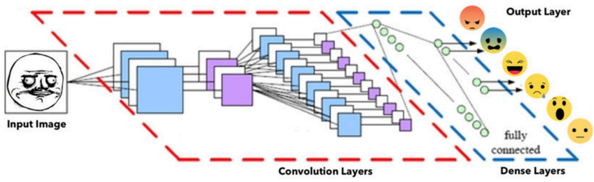
The gathered feature points are processed to prepare the inputs for the neural network. Primary inputs for the neural network are:

i. *Eye Height* = $[(c4 - c3) + (d4 - d3)]/2$
ii. *Eye Width* = $[(c2 - c1) + (d2 - d1)]/2$
iii. *Mouth Height* = $(f4 - f3)$
iv. *Mouth Width* = $(f2 - f1)$
v. *Eyebrow to Eye center height* $= [(c5 - a1) + (d5 - b1)]/2$
vi. *Eye center to Mouth center height* $= [(f5 - c5) + (f5 - d5)]/2$
vii. *Left eye center to Mouth top corner length*
viii. *Left eye center to Mouth bottom corner length*
ix. *Right eye center to Mouth top corner length*
x. *Right eye center to Mouth bottom corner length* [2, 18-24]

The neural network has been trained to recognize the emotions such as happiness, sadness, anger, disgust, surprise, fear and neutral expression using the obtained datapoints. Images from the FER-2013 Database are used to train the network.

III. Algorithms Involved





In order to implement the algorithm involved in this project various libraries of python have been used which have been explained below:

```
#importing import libraries that are going to be used
#in the Facial Emotions Recognition using WebCam
from keras.preprocessing.image import img_to_array
import imutils
import cv2
from keras.models import load_model
import numpy as np
import pandas as pd
import os
import matplotlib.pyplot as plt
import tensorflow.keras as tk
```

1. *The OS module provides functions for the interaction with the operating system.*
2. *NumPy is an important library containing mathematical functions related to arrays.*
3. *Matplotlib.pyplot is a plotting library with the help of which we will be plotting our final image and graphs.*
4. *OpenCV provides a wide range of tools for image and video processing, computer vision tasks, and machine learning.*
5. *Tensorflow.keras in python makes it easy to make Convolutional Neural Network. Keras.layer input is the starting tensor and allows us to build a keras model just by knowing the inputs and outputs of the model.*
6. *The imutils library is a collection of convenience functions for OpenCV in Python. It simplifies common tasks such as resizing, rotating, and displaying images.*

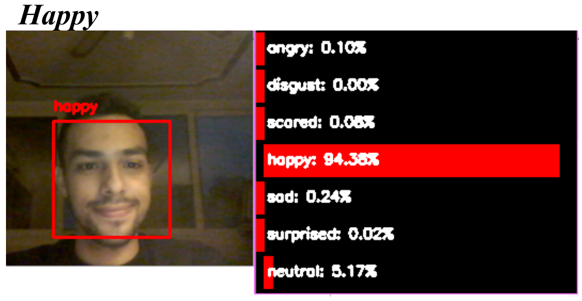
Deep neural networks consist of input layer, hidden layers, and activation function. The input layer receives the initial input data, upon which it is passed through many hidden layers. Neurons in hidden layers perform computations based on the input and weights, generating increasingly abstract and hierarchical representations of the data. Each layer is responsible for identifying patterns in the images through 19 characteristic points. These points are in eyes, mouth, eyebrows, and noses. And the final output is a number that represents any one of the emotions.

A very good way to use DL is to categorize images to create CNN. Convolutional layers use filters to find patterns like edge or corner in the image. We can detect more sophisticated objects or patterns like eyes, hair, etc. by making the network deeper. A filter is a small matrix or kernel where you can specify the size ($n \times n$). Each time when the input is given, filter will slide over every ($n \times n$) block in the image. It repeats the same process until all pixels in the image are covered.

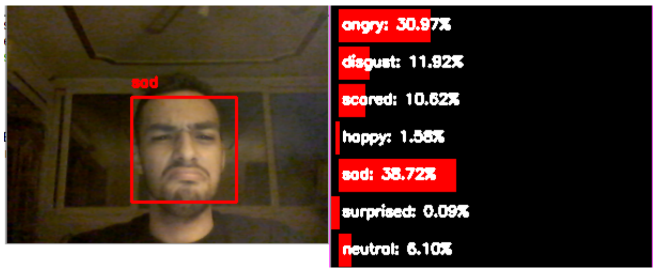
IV. Results

Thus, using the supplied parameters, we train our model on standard datasets. Our model follows these steps through the Algorithms listed above and classifies images as:

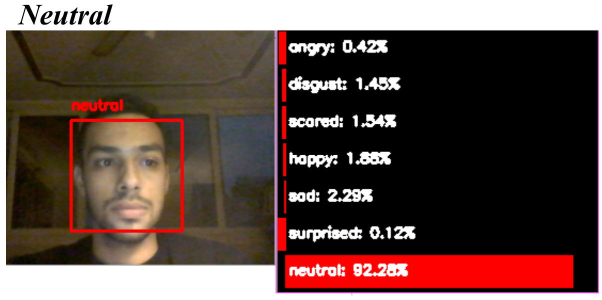
1. ***Happy***



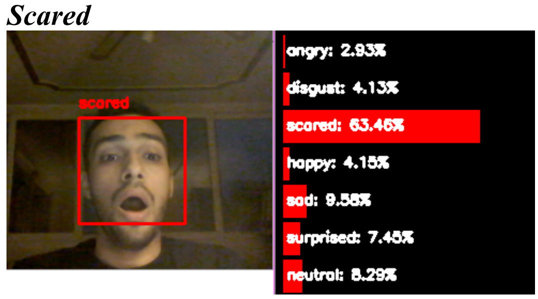
2. ***Sad***



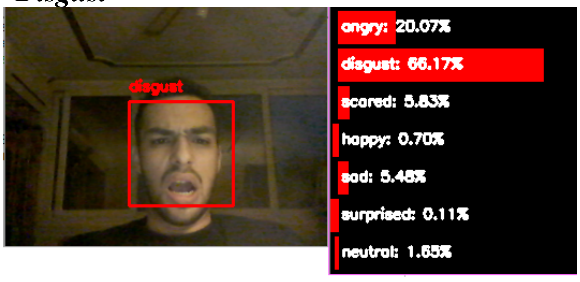
3. ***Neutral***



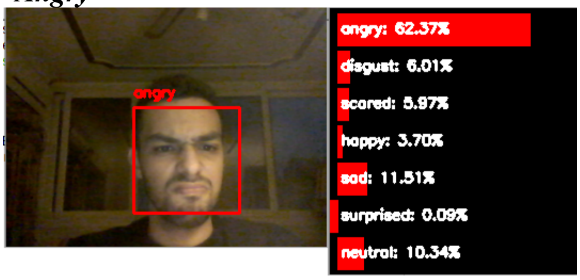
4. ***Scared***



5. ***Disgust***



6. ***Angry***



V. Novel Applications of Emotion Recognition

1. ***Driver Distraction Detection***
Facial emotion recognition can be a valuable component in driver distraction detection systems. The goal of such systems is to monitor the driver's behavior and identify situations where their attention may be diverted, or their emotional state may impact their ability to drive safely. Here's how facial emotion recognition can be applied in driver distraction detection:
   - i. **Real-time Monitoring:** Facial emotion recognition systems continuously analyze the driver's facial expressions in real-time using a camera. The system detects facial features and tracks changes in expressions.
   - ii. **Emotion Classification:** The system classifies the driver's emotional state based on their facial expressions. Common emotions of interest include happiness, sadness, anger, and surprise.
   - iii. **Distraction Detection:** Facial emotion recognition can be used to identify signs of distraction or stress in the driver's facial expressions. Sudden changes in emotions or expressions inconsistent with driving conditions may indicate distraction.
   - iv. **Integration with Other Sensors:** Facial emotion recognition can be integrated with other sensors, such as eye-tracking systems and vehicle sensors. Combining data from multiple sources enhances the accuracy of distraction detection.
   - v. **Alerts and Warnings:** When the system detects signs of distraction or emotional distress, it can trigger alerts or warnings to bring the driver's attention back to the road. Alerts can be visual, auditory, or haptic, depending on the severity of distraction.
   - vi. **Personalized Responses:** The system may be designed to provide personalized responses based on the driver's emotional state. For example, if stress is detected, the system might suggest taking a break or providing calming notifications.

2. ***Facial emotion recognition for item ratings in shopping malls***
Implementing facial emotion recognition for item ratings in shopping malls can create an engaging and dynamic shopping environment, providing valuable insights for both customers and mall management. Additionally, it adds a layer of personalization to the shopping experience, making it more enjoyable and tailored to individual preferences.
   - i. **Facial Emotion Capture:** Deploy cameras or smart devices with cameras at strategic locations within the shopping mall. Capture customers' facial expressions as they interact with products or view displays.
   - ii. **Real-Time Emotion Analysis:** Use facial emotion recognition algorithms to analyze real-time facial expressions. Identify emotions such as happiness, surprise, satisfaction, or dissatisfaction.
   - iii. **Dynamic Product Ratings:** Generate dynamic and real-time ratings for products based on the collective emotional responses of customers. Emotion intensity and frequency can influence the product ratings.
   - iv. **User Engagement:** Encourage customers to actively participate by expressing their emotions through facial expressions. Provide incentives, such as discounts or loyalty points, for customers who contribute to the emotional rating system.
   - v. **Data Analytics and Insights:** Aggregate and analyze the collected emotional data over time. Identify trends, popular products, and areas with high emotional engagement.

3. ***Facial emotion recognition for online advertisement recommendation***
Integrating facial emotion recognition into online ad recommendation systems adds a layer of personalization and responsiveness, creating a more engaging and enjoyable user experience. It also provides advertisers with valuable insights to refine their ad strategies. However, it's **crucial to handle user privacy** and data ethics responsibly throughout the implementation.
   - i. **User Engagement Analysis:** Implement facial emotion recognition to analyze users' emotional responses while interacting with online content, including ads. Capture facial expressions to determine emotional engagement levels.
   - ii. **Ad Placement Optimization:** Optimize the placement of ads on websites or platforms based on users' emotional engagement. Prioritize ad positions that historically generate higher emotional responses.
   - iii. **Cross-Platform Integration:** Extend facial emotion recognition across multiple platforms and devices for a seamless and consistent ad experience. Synchronize

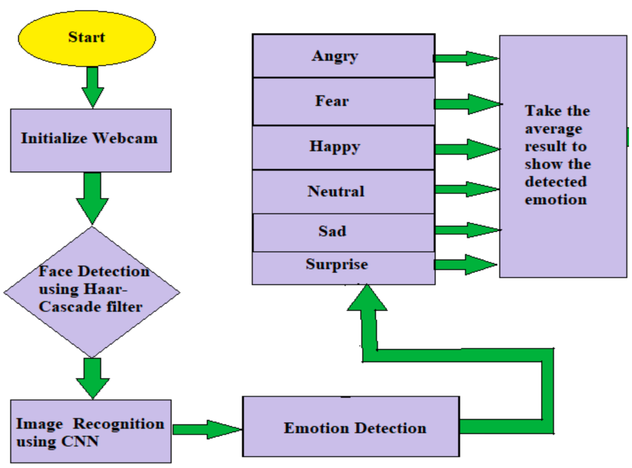
emotional data to provide cohesive recommendations.

iv. **Privacy Measures:** Prioritize user privacy by clearly communicating the use of facial emotion recognition for ad recommendations. Allow users to opt-in or opt-out of emotion-based ad targeting.

VI. Issues and Challenges

1. **Privacy and Ethical Concerns:** The project acknowledges the importance of addressing privacy and ethical considerations related to facial data collection. Clear user consent mechanisms and anonymization techniques have been implemented to mitigate these concerns.
2. **Generalization:** Achieving robust generalization across diverse demographics and cultural contexts remains an ongoing challenge, emphasizing the need for continuous refinement and inclusivity in model training.
3. **Future Directions:** The success of the Facial Emotion Recognition project opens avenues for future exploration and improvement:
4. **Fine-tuning and Optimization:** Further refinement of the model through fine-tuning and optimization could enhance accuracy and performance.
5. **Incorporating Multimodal Data:** Integration with other modalities, such as voice or physiological signals, could provide a more comprehensive understanding of emotional states.
6. **Adaptive Learning:** Implementing adaptive learning mechanisms to dynamically adjust to evolving user behaviors and preferences.
7. **Extended Applications:** Exploring additional applications in emerging fields like augmented reality, emotional gaming, and affective computing.

VII. Summary



Flowchart for emotion detection from Live Camera Feed

Our project has successfully demonstrated the effectiveness of leveraging advanced computer vision techniques to analyze and interpret human emotions through facial expressions. The primary objectives of the project were to design and implement a robust emotion recognition system, explore its applications in various domains, and assess its impact on user experiences.

Some of the key findings of this projects are:

i. **Accuracy and Performance:** The developed facial emotion recognition model exhibited commendable accuracy in detecting a range of emotions, including happiness, sadness, anger, surprise, and more. Rigorous testing and validation procedures ensured reliable performance across diverse datasets.

ii. **Real-time Analysis:** The system demonstrated real-time capabilities, making it suitable for applications requiring instantaneous feedback, such as interactive installations, virtual environments, and dynamic content personalization.

iii. **Applications in Diverse Sectors:**

   a. ***Human-Computer Interaction:*** *The technology has promising applications in human-computer interaction, enabling more intuitive and responsive user interfaces.*

   b. ***Healthcare:*** *Facial emotion recognition can contribute to mental health assessments, offering potential insights into emotional well-being.*

   c. ***Education:*** *Integration into educational tools enhances understanding of student engagement and emotional states during learning activities.*

iv. **User Experience Enhancement:** The incorporation of facial emotion recognition in various scenarios has the potential to significantly enhance user experiences. Personalized recommendations, dynamic content adaptation, and emotion-driven interactions contribute to a more engaging and tailored user journey.

VIII. References

[1] Pise, Anil Audumbar, Mejdal A. Alqahtani, Priti Verma, Dimitrios A. Karras, and Awal Halifa. "Methods for Facial Expression Recognition with Applications in Challenging Situations." Computational Intelligence and Neuroscience 2022 (2022).

[2] Widanagamaachchi, W. N., and A. T. Dharmaratne. "Emotion recognition with image processing and neural networks." Research Gate (2009).

[3] P. S Aleksic and A. K. Katsaggelos, “Automatic facial ex- pression recognition using facial animation parameters and multistream hmms,” IEEE Transactions on Information Forensics and Security, vol. 1, no. 1, pp. 3–11, 2006.

[4] H. Chouhayebi, J. Riffi, M. A. Mahraz, Y. Ali, and T. Hamid, “Facial expression recognition using machine learning,” in Proceedings of the 2021 Fifth International Conference On Intelligent Computing in

Data Sciences (ICDS), pp. 1–6, IEEE, Seoul, Korea, November 2021.

[5] Anil. Audumbar Pise, H. Vadapalli, and I. Sanders, “Relational reasoning using neural networks: a survey,” Inter- national Journal of Uncertainty, Fuzziness and Knowledge- Based Systems, vol. 29, pp. 237–258, 2021.

[6] Puri, Archit Gupta, Manas Sikri, “ Emotion Detection using Image Processing in Python,” in IEEE Conference ID: 42835 2018 5th International Conference, 14th - 16th March 2018.

[7] Ninad Mehendale, “Facial emotion recognition using convolutional neural networks (FERC),” 18 February 2020.

[8] James Pao, “Emotion Detection Through Facial Feature Recognition,” in International Journal of Multimedia and Ubiquitous Engineering, November 2017.

[9] Dan Duncan, Gautam Shine, Chris English, "Facial Emotion Recognition in Real-Time," November 2016.

[10] Shivam Gupta, “Facial emotion recognition in real-time and static images,” in 2nd International Conference on Inventive Systems and Control (ICISC) IEEE, 28 June 2018.

[11] Aneta Kartali, Miloš Roglić, Marko Barjaktarović, Milica Đurić-Jovičić, Milica M. Janković, “Real-time Algorithms for Facial Emotion Recognition: A Comparison of Different Approaches,” in 2018 14th Symposium on Neural Networks and Applications (NEURAL), Nov 2018.

[12] Jonathan, Andreas Pangestu Lim, Paoline, Gede Putra Kusuma, Amalia Zahra, “Facial Emotion Recognition Using Computer Vision,” in Indonesian Association for Pattern Recognition International Conference (INAPR) IEEE, 31 January 2019.

[13] Renuka S. Deshmukh, Vandana Jagtap, Shilpa Paygude, “Facial emotion recognition system through machine learning approach,” in International Conference on Intelligent Computing and Control Systems (ICICCS) IEEE, 11 January 2018.

[14] Ninad Mehendale “Facial emotion recognition using convolutional neural networks.

[15] Ali, Md. Forhad & Khatun, Mehenag & Turzo, Nakib. (2021). Facial Expression Recognition Using Neural Network.

[16] Byoung Chul Ko, “A Brief Review of Facial Emotion Recognition Based on Visual Information,” 30 January 2018

[17] Brunelli, R., and Poggio., T., Face recognition: features versus templates. IEEE Transactions on Pattern Analysis and Machine Intelligence, 15(10):1042–1052, 1993.

[18] Dumasm, M., Emotional expression recognition using support vector machines. Technical report,Machine Perception Lab, Univeristy of California, 2001.

[19] Lisetti, C. L., and Rumelhart, D. E., Facial expression recognition using a neural network. In Proceedings of the Eleveth International FLAIRS Conference. Menlo Park, pages 328– 332. AAAI Press, 1998.

[20] Nilsson, M., Nordberg, J., and Claesson, I., Face detection using local smqt features and split up snow classifier. In IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2007.

[21] Pantic, M., Rothkrantz, L. J. M., and Koppelaar, H., Automation of nonverbal communication of facial expressions. In: EUROMEDIA 98, SCS International, pages 86–93, 1998.

[22] Sebe, N., Sun, Y., Bakker, E., Lew, M. S., Cohen, I., and Huang, T. S., Towards authentic emotion recognition.

[23] Wallhoff, F., Facial expressions and emotion database, last accessed date: 01st june 2009, 2006.

[24] Yang, M. H., Ahuja, N., and Kriegman, D., A survey on face detection methods, 1999.

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